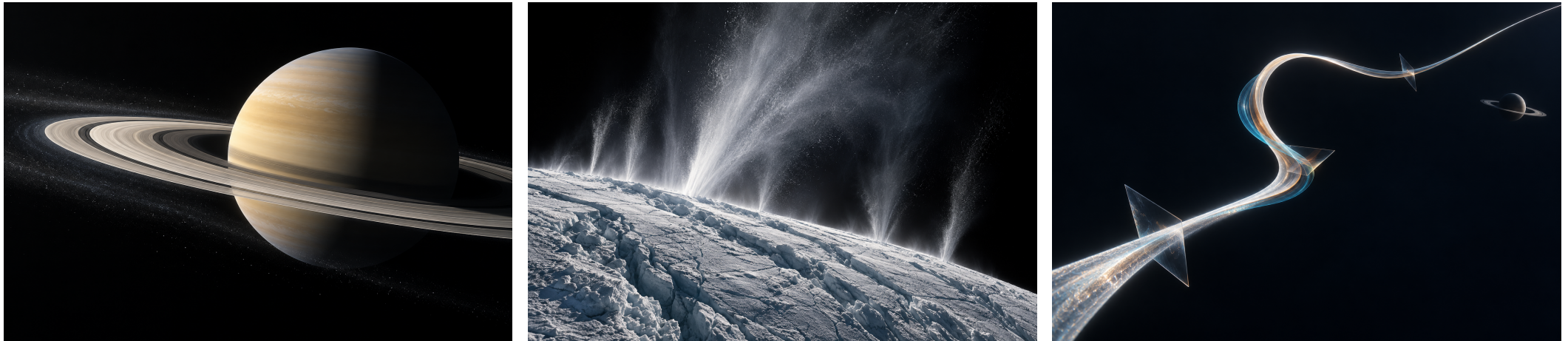


ASTRA SPPT SCM Visual Atlas

Original diagrams and illustrations for the physics research companion



Six scientific figures · Two analytic animations · Three generated illustrations

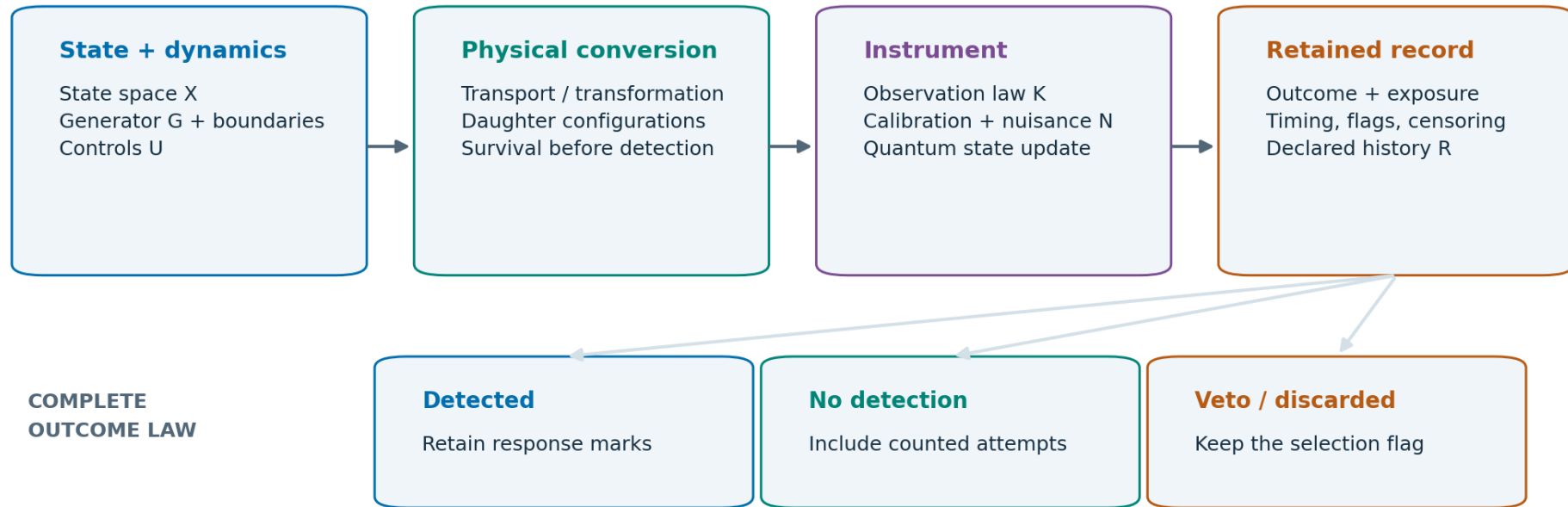
The plots show conditional mathematics and synthetic display choices. The generated art is conceptual. None of these visuals supplies experimental validation, a common microscopic law, a spacecraft measurement or a replacement for the stable SPPT and ASTRA v1.0.7 science authority.

Animation pages show still previews. Full motion is supplied as GIF and MP4 files in the accompanying bundle. Detailed captions, parameter choices, source links and exact generation prompts accompany the assets.

01 / SCIENTIFIC ATLAS

A complete route from source to record

A shared inference contract; physical generators, units and likelihoods remain domain-specific.



Classical kernels $P_u(dy) = \int K_u(dy | z, v) C_u(dz | x, v) r(dx)$

Quantum instrument $P_u(y | \rho) = \text{Tr } \mathcal{I}_{y,u}(\rho)$, $\sum_y \mathcal{I}_{y,u}$ is trace preserving

Schematic. Define the attempt denominator before adding no-detection outcomes; missing coverage is not a negative result. Rejection criterion V and nuisance support must be declared for each model. Source: research companion, Section 2, Eqs. (1)-(2).

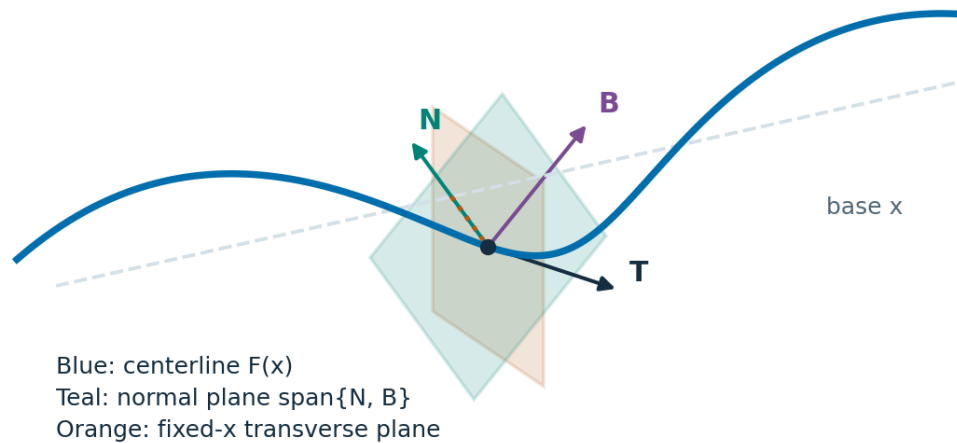
A common record-keeping discipline with distinct physical models. Include no-detection and veto outcomes only for operationally counted attempts. Missing coverage is not a negative result.

Files: scientific/figures/01_complete_observation_contract.svg, scientific/figures/01_complete_observation_contract.png

A complex graph and its normal plane

The two transverse graph coordinates and the geometric normal frame are distinct constructions.

Orthographic view of an analytic helix



EXACT ILLUSTRATIVE GEOMETRY

$$F(x) = (x, A \cos kx, A \sin kx)$$

$$T = F_x / |F_x|, \quad B = T \times N$$

$$N = (0, -\cos kx, -\sin kx)$$

N and B span the plane perpendicular to the tangent T .

The fixed- x plane instead holds the complex graph height $U = a(\text{Re } \psi, \text{Im } \psi)$.

Finite slope requires a tangential reparametrization in graph flow.

Analytic geometry, not a time-evolved PDE solution or a model of matter. $A = 0.6 L_0$, $k = 1/L_0$; arbitrary length L_0 . The normal frame does not establish spin, charge or a gravitational metric. Sources: report Section 3; Hasimoto (1972).

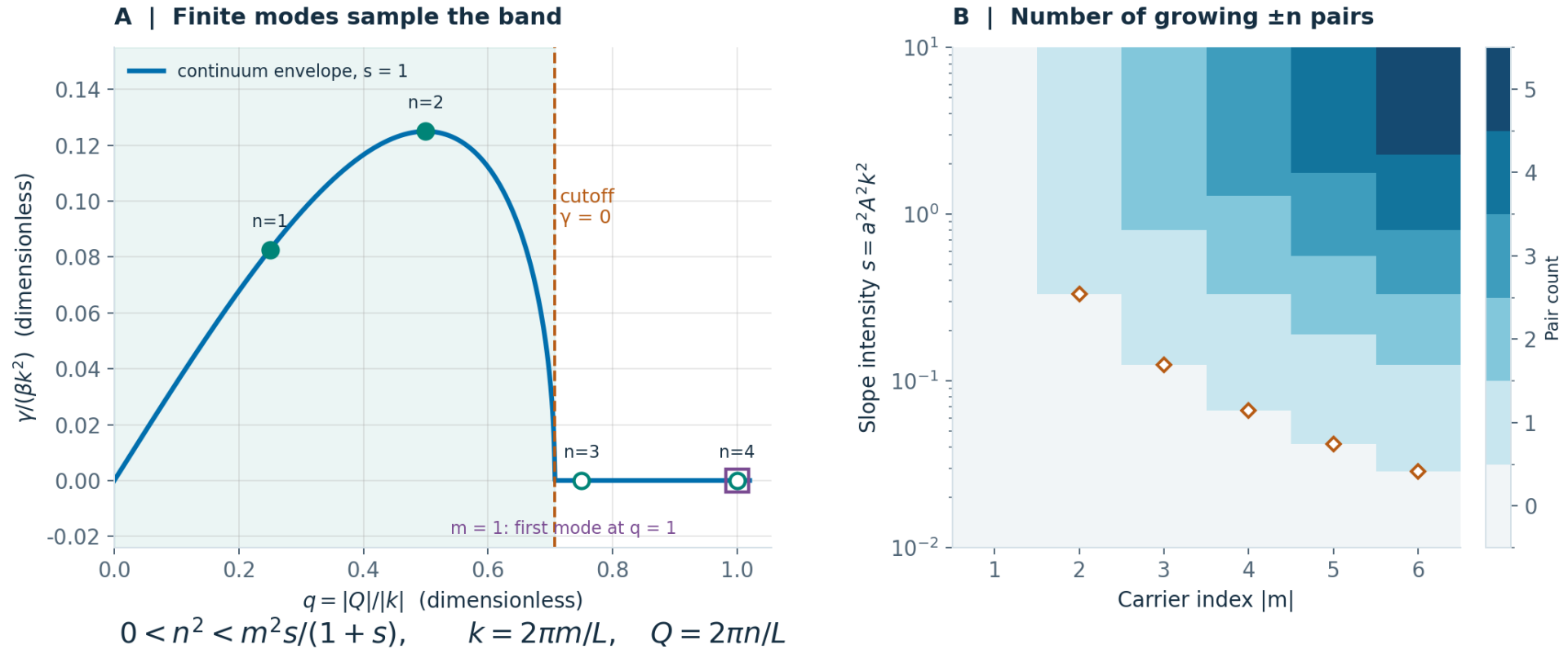
Exact illustrative helix geometry. The fixed- x graph plane and the plane normal to the tangent generally differ at finite slope. This is neither a PDE solution nor a model of matter.

Files: scientific/figures/02_complex_graph_normal_plane.svg, scientific/figures/02_complex_graph_normal_plane.png

03 / SCIENTIFIC ATLAS

A growth band is not a stability verdict

Exact linearized predictions for a specified periodic WKI carrier; all plotted inputs are synthetic.



Diamonds mark the first-mode threshold. Equality is marginal: a nonzero cutoff can have Jordan-block secular growth.

No growing mode is not proof of nonlinear stability. β has units L_0^2/T_0 ; rates are scaled by βk^2 . Left: $s = 1$, $m = 4$.

Right: exact integer count on each plotted grid cell. Sources: report Section 4, Eqs. (15)-(23); Zhang et al. (2019) for the WKI equation.

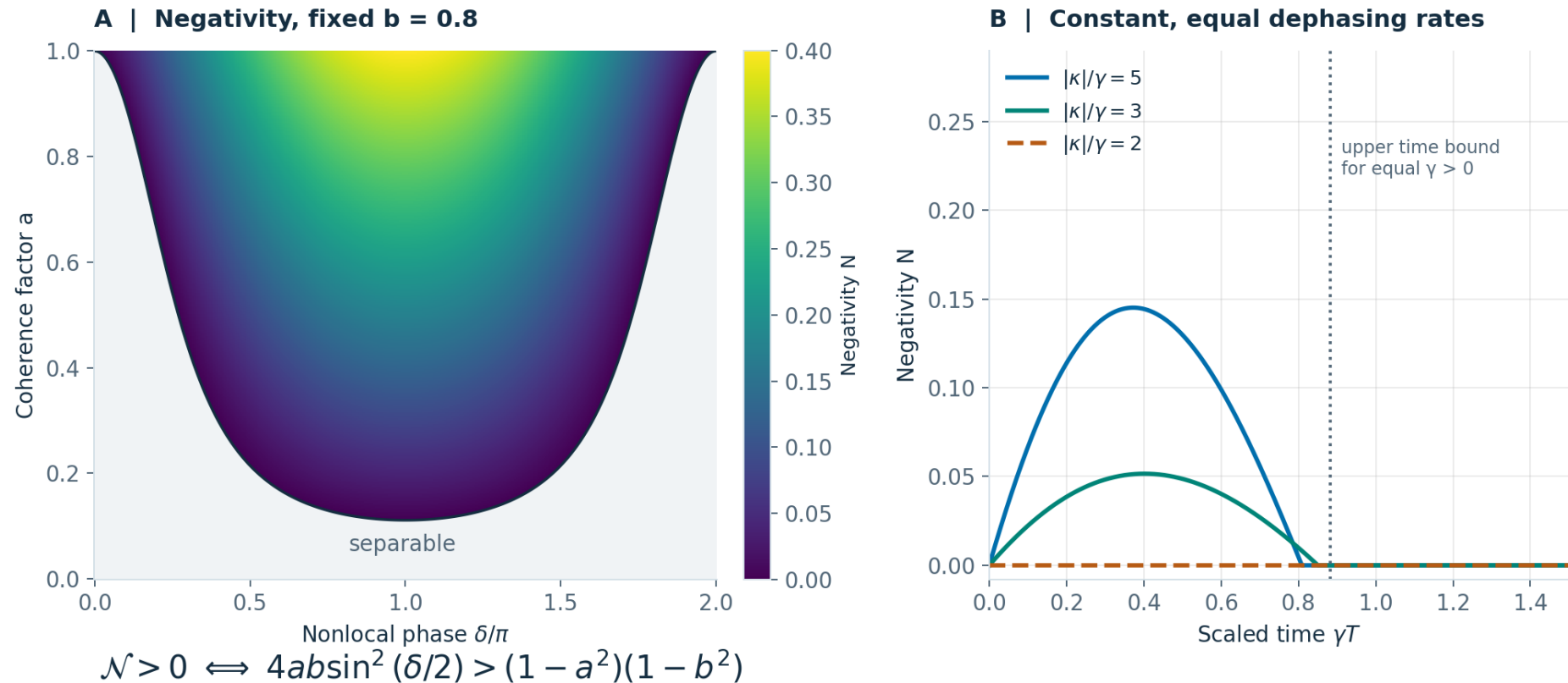
Synthetic, exact linearized WKI predictions. At $s = 1$ and $m = 4$, the $n = 1$ and $n = 2$ pairs grow exponentially. No growing mode does not prove nonlinear stability; marginal modes require separate analysis.

Files: scientific/figures/03_wki_discrete_growth_bands.svg, scientific/figures/03_wki_discrete_growth_bands.png

04 / SCIENTIFIC ATLAS

Entanglement needs phase and surviving coherence

A two-qubit model map, not an observed gravity signal or a new entanglement formula.



Balanced path qubits; deterministic diagonal interaction; independent local Z dephasing. For panel B: $a = b = \exp(-\gamma T)$, $\delta = \kappa T$.

Synthetic parameter scan. No apparatus sensitivity, witness uncertainty or gravitational-origin inference is modeled.

Spectrum prior art: Rizaldy et al., arXiv:2609.10697v1 (2026), Eqs. 28-30 and 61-62. Report Section 5 supplies the rate consequence.

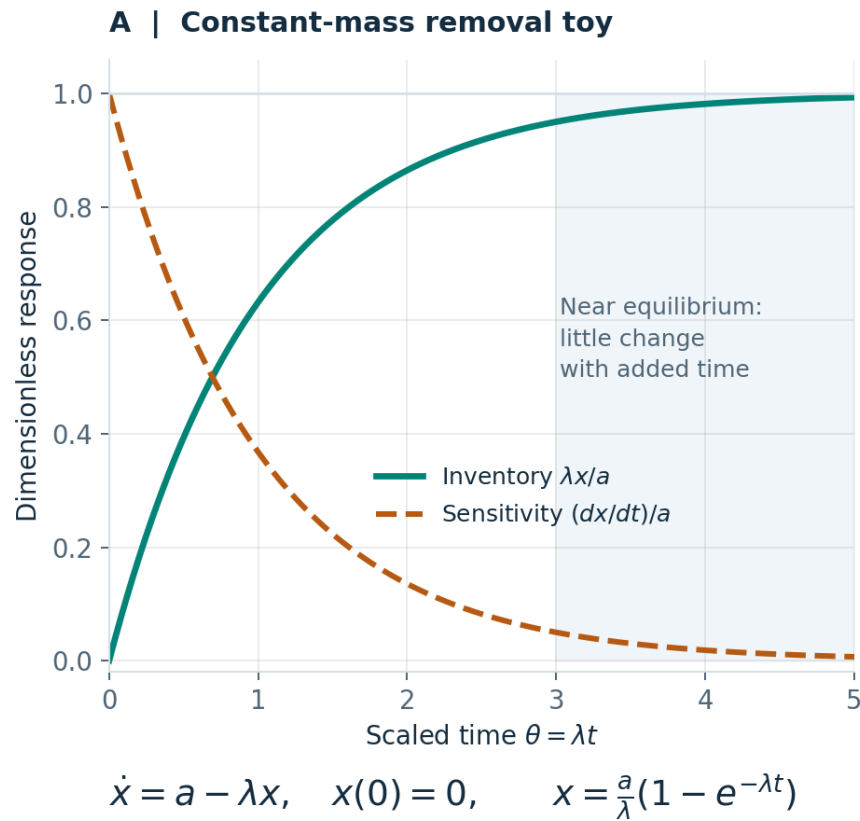
An original plot of an established two-qubit result. Independent local Z dephasing and a deterministic nonlocal phase are assumed. This is not an observed gravity signal, and no physical apparatus is simulated.

Files: scientific/figures/04_phase_coherence_entanglement.svg, scientific/figures/04_phase_coherence_entanglement.png

05 / SCIENTIFIC ATLAS

A clean ring is not a unique formation clock

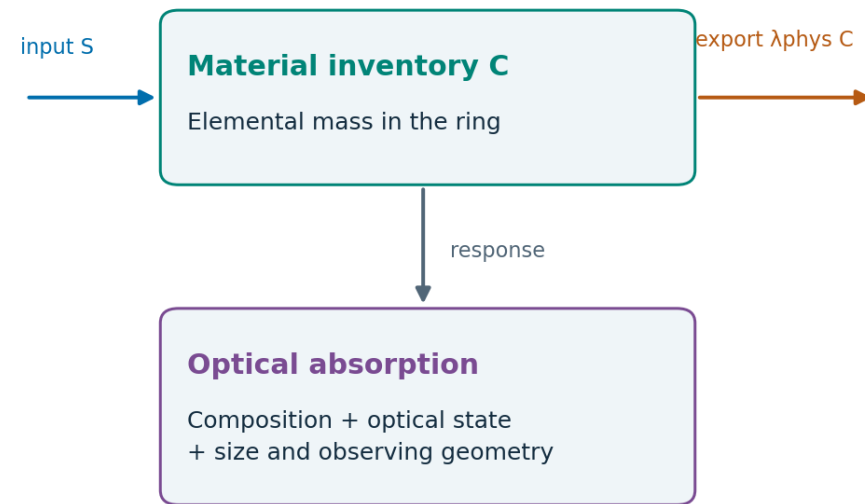
A saturating inventory loses time sensitivity; optical alteration and physical export need separate measurements.



Toy assumptions: constant total ring mass, input and physical removal; initially clean, dilute inventory. No ring age is fitted.

Here a and λ have units $1/\text{time}$; x is a mass fraction. An all-time literal fraction requires $a/\lambda \leq 1$ (or a restricted dilute regime).
Sources: report Appendix C2, Eqs. P1-P2; Ricerchi & Crida (2026). The plotted toy is not their complete ring-evolution model.

B | Keep both ledgers



Altered absorption need not remove an element.

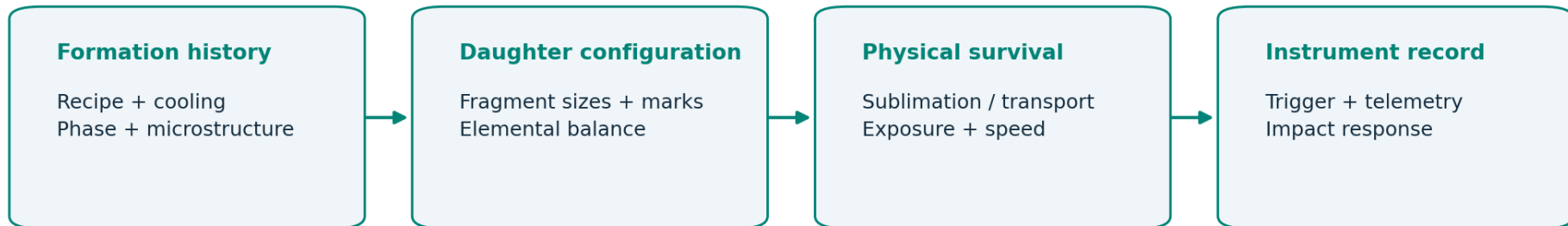
A constant-mass removal toy, not a Saturn ring-age fit. Near equilibrium, inventory changes carry little age information. Optical alteration and physical material export require separate measurements.

Files: scientific/figures/05_ring_inventory_optical_clock.svg, scientific/figures/05_ring_inventory_optical_clock.png

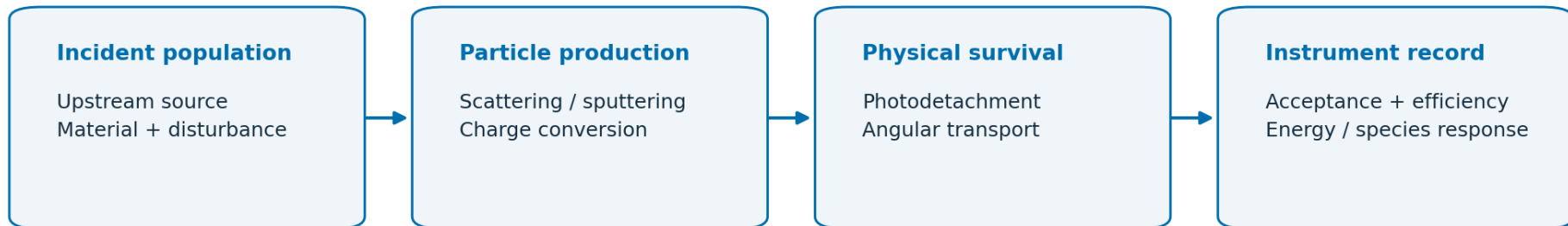
Formation is only the start of a detected sample

Enceladus grains and lunar ions share an inference discipline, with different physics at every stage.

ENCELADUS



LUNAR SURFACE



Control the Enceladus comparison

Measure chemistry and phase before and after drying.
Compare the same grain, or a declared matched population.

Control the lunar comparison

Separate ionization probability from detector calibration.
Retain source exposure, angular support and non-detections.

Proposed experimental / inference design, not a Cassini or lunar reconstruction. Physical loss precedes detector selection.
Sources: report Sections 6 and C4-C6; Postberg et al. (2026), EPSC2026-1326, and the Chang'e-6 studies (2025-2026).

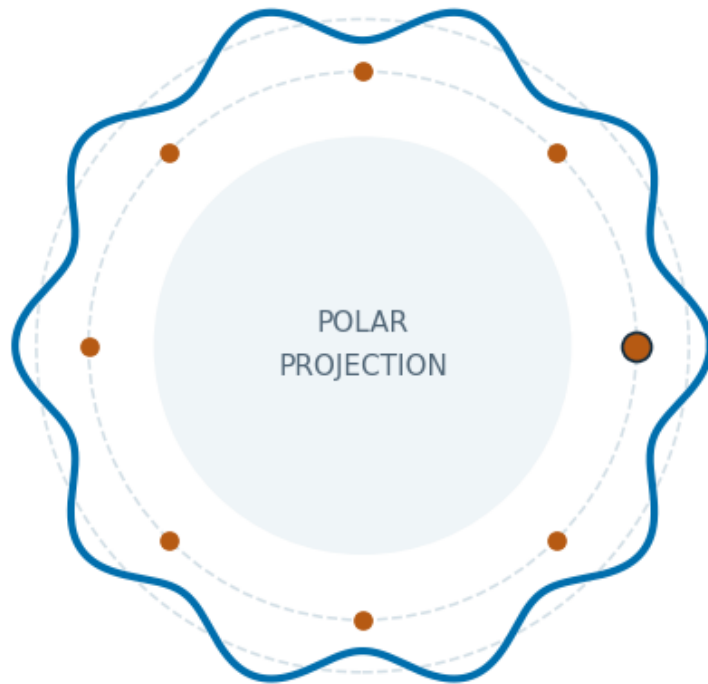
Proposed source-to-record comparisons. Formation, physical survival and detector selection remain distinct. The two rows do not share a physical kernel, and the diagram makes no crystallographic claim for Cassini CAT.

Files: scientific/figures/06_planetary_transfer_comparison.svg, scientific/figures/06_planetary_transfer_comparison.png

A1 / SCIENTIFIC ATLAS

Pattern speed is not material speed

A synthetic ten-lobed wave and prescribed tracers, shown from above. No Saturn speed is fitted.

**BLUE | wave pattern**

$$r = 1 + \epsilon \cos[10(\theta - \Omega_p t)]$$

One wavelength advances per loop.

ORANGE | tracers

$$\theta_j(t) = \theta_j(0) + \Omega_m t$$

Two circuits per loop at fixed radius.

These motions were prescribed independently. No advection or shallow-water equations were solved.

Kinematic illustration only. Radius and angular rates are arbitrary; not a spacecraft image or a reconstruction.

Motivation: report Appendix C3 and ESA/Hubble's 2 September 2026 Saturn-decagon release.

Animation preview. A synthetic ten-lobed pattern advances one wavelength per loop while prescribed tracers complete two circuits. Display rates are arbitrary. No fluid or advection equations were solved.

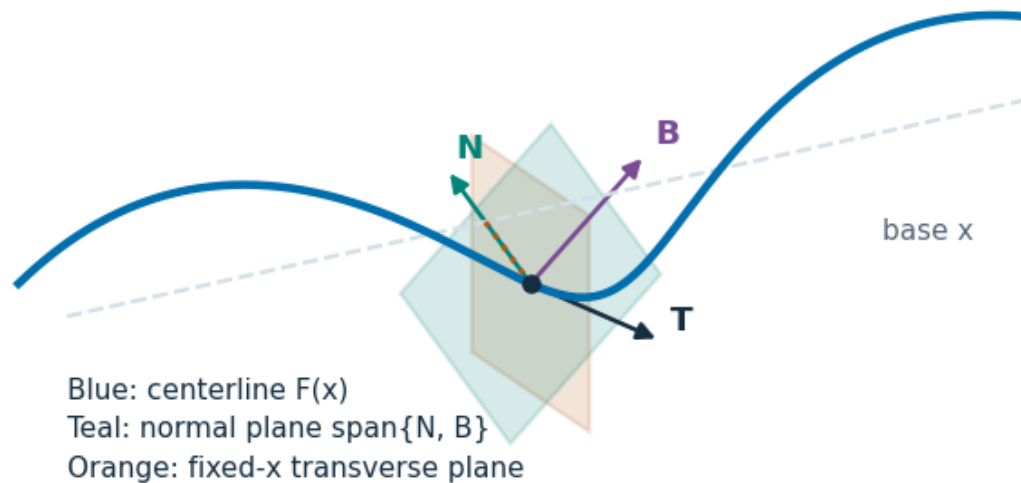
Files: scientific/animations/A1_pattern_vs_material_motion.gif, scientific/animations/A1_pattern_vs_material_motion_poster.png, scientific/animations/A1_pattern_vs_material_motion.mp4

A2 / SCIENTIFIC ATLAS

A bending graph and its moving normal plane

Analytic geometric interpolation; displayed time is a drawing parameter, not physical evolution.

Orthographic view of an analytic helix



BLUE | centerline

$$F(x, t) = (x, A(t) \cos kx, A(t) \sin kx)$$

TEAL | normal plane

The frame is recomputed from the tangent at each step.

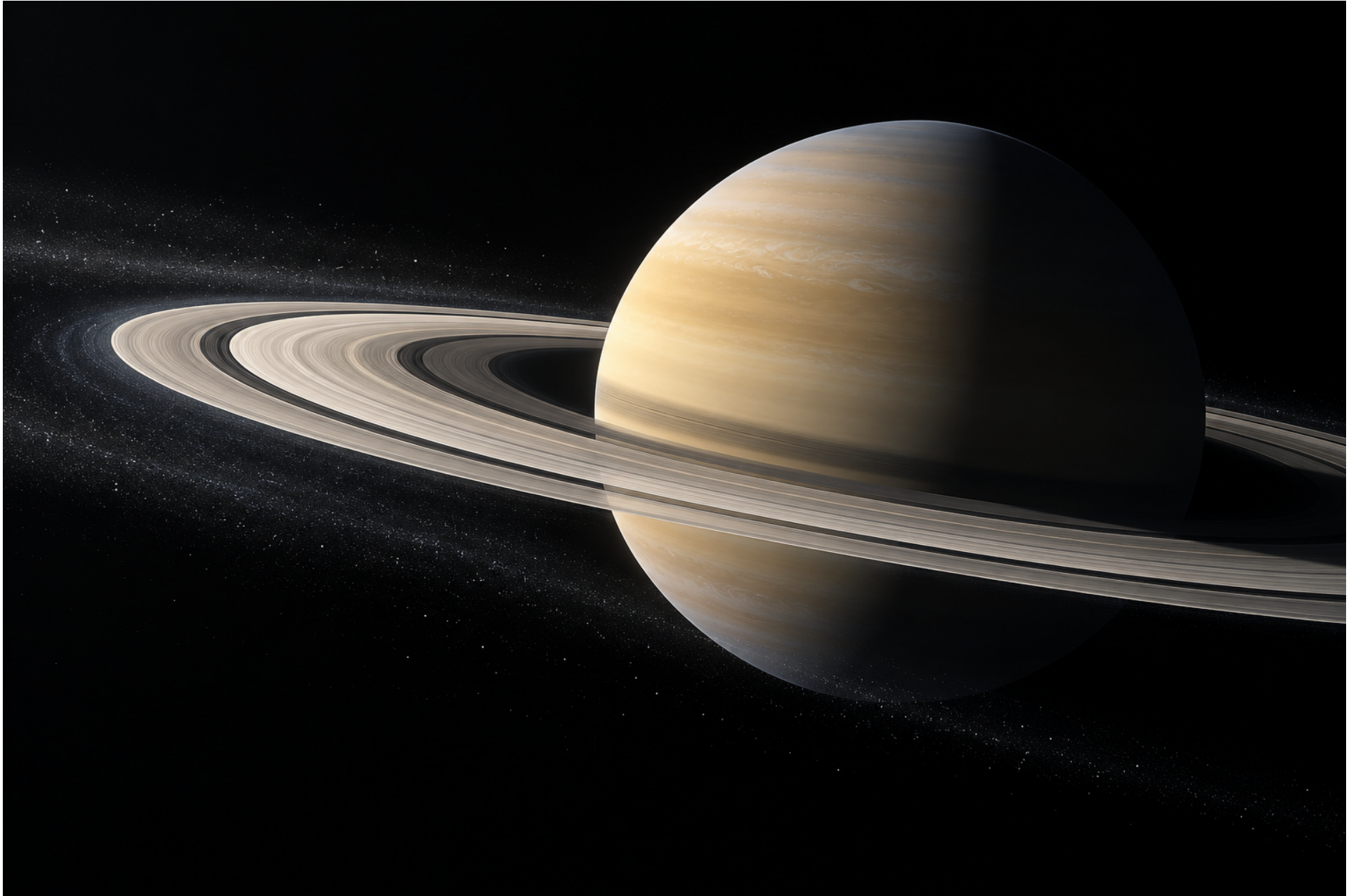
ORANGE | fixed-x plane

Finite slope keeps graph and normal planes distinct.

$A(t)/L_0 = 0.55 + 0.15 \cos(2\pi t/8)$; $k = 1/L_0$. Length and display time are arbitrary. No PDE is integrated.
 T, N and B are orthonormal analytically in every frame. This is geometry, not a microscopic material mechanism.

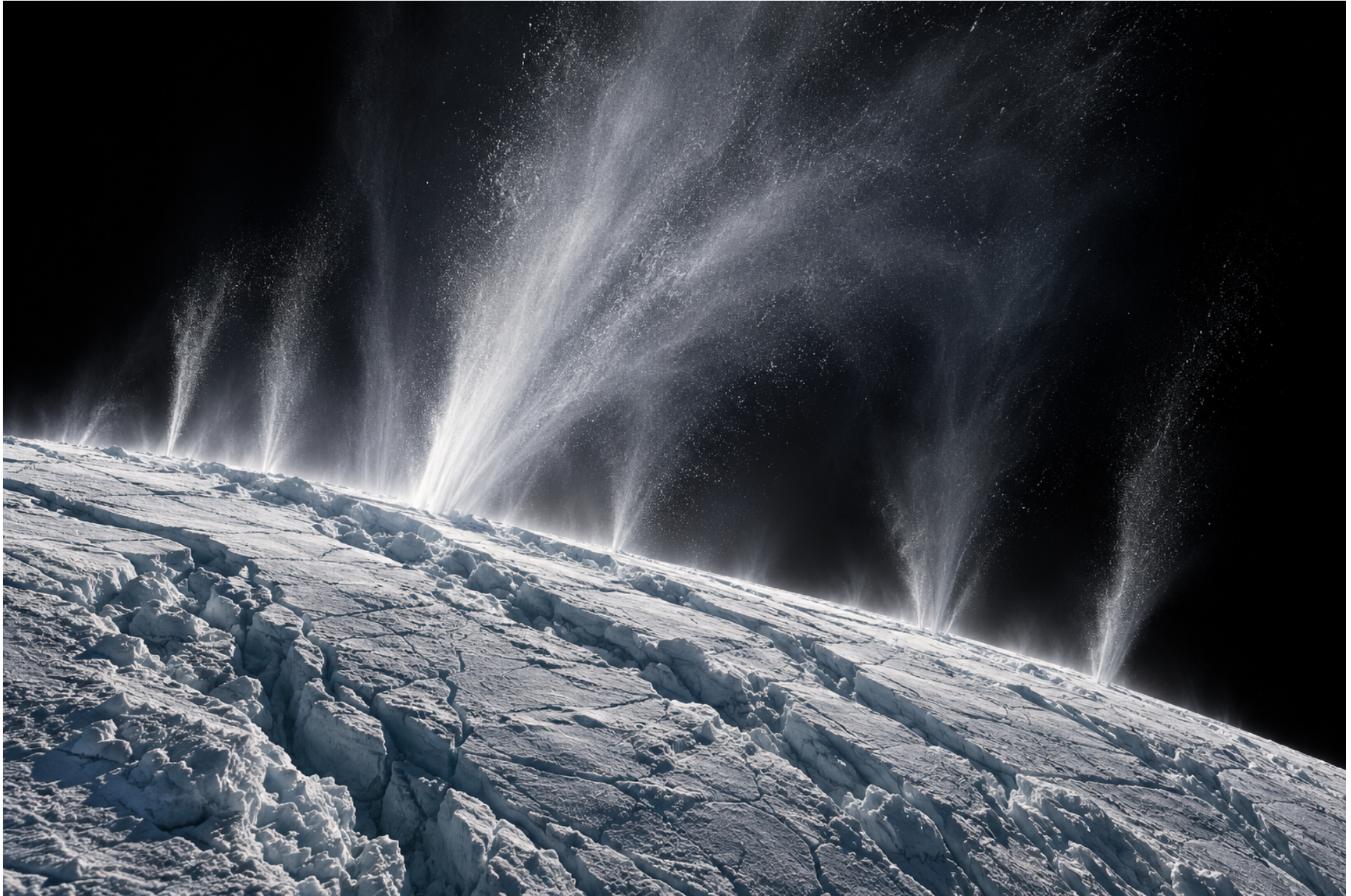
Animation preview. The normal frame is recomputed through a prescribed helix family. Display time is an interpolation parameter. This is not integrated SCM, WKI, binormal-flow or quantum dynamics.

Files: scientific/animations/A2_bending_graph_normal_frame.gif, scientific/animations/A2_bending_graph_normal_frame_poster.png, scientific/animations/A2_bending_graph_normal_frame.mp4



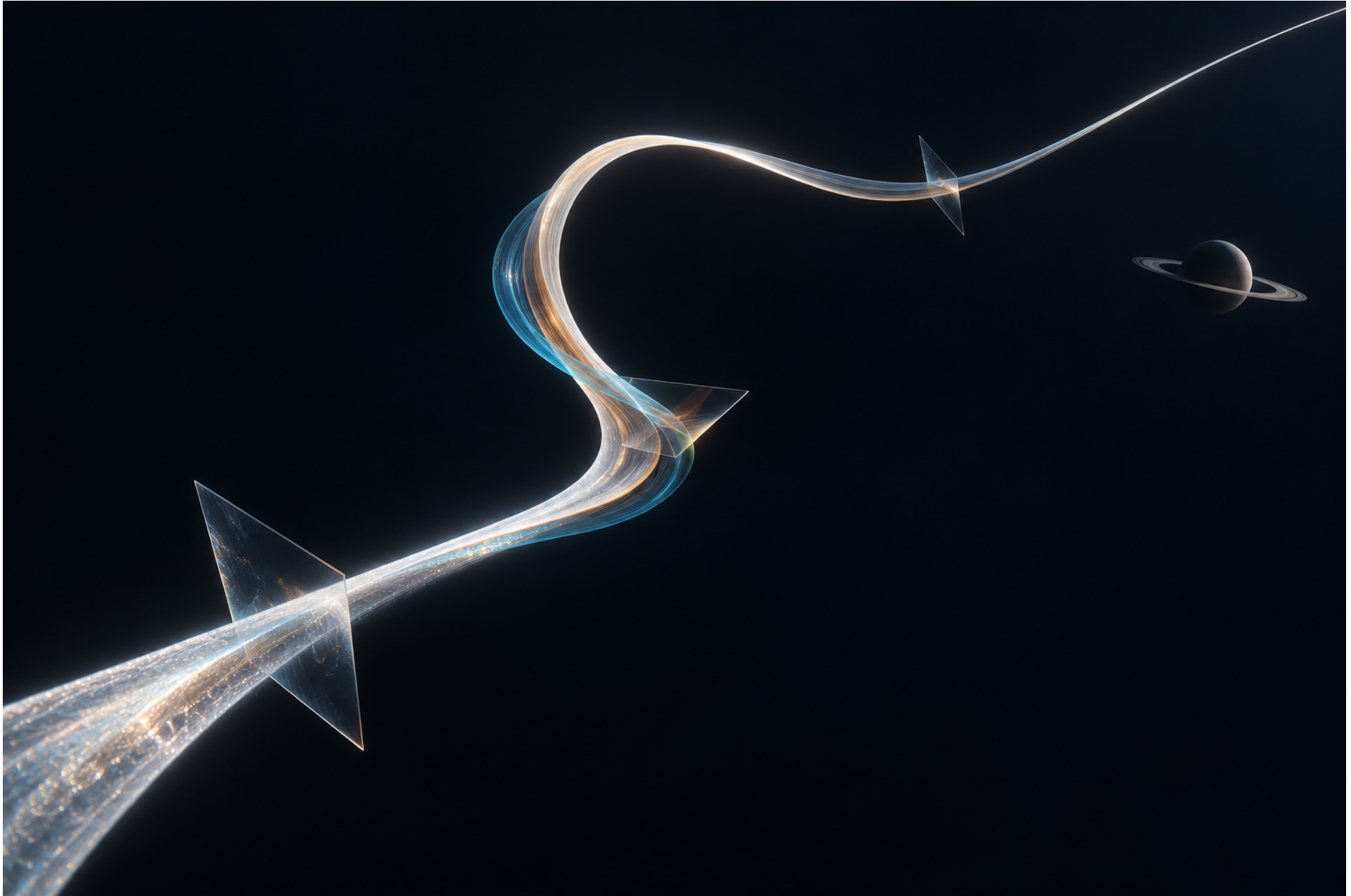
AI-generated illustration of Saturn's main rings with an artistically enhanced off-plane dust motif. Dust brightness, particle size and spatial distribution are illustrative, not measured; this is not a spacecraft image. The icy main-ring setting is physically informed. The visible dust is a compositional device, not a reconstruction of a particular silicate population, a measured halo boundary, or a demonstration of transport or ring age.

Files: illustrations/images/01-saturn-ring-dust-illustration.png



AI-generated illustration inspired by Enceladus's south-polar ice fractures and plumes. Terrain, jets, viewing geometry and scale are artistic; this is not a Cassini image or a measured reconstruction. Water-ice and vapor plumes from south-polar fissures motivate the scene. Individual fissures, jet locations and grain sizes are invented for the illustration. The image does not reveal subsurface structure, freezing history, ocean chemistry or biological activity.

Files: illustrations/images/02-enceladus-plume-illustration.png



AI-generated speculative concept art combining a bent-filament and normal-plane motif with a separate Saturn-like echo. It represents no measured field, computed solution or experimentally established connection. The ribbons and plane fragments are artistic motifs rather than a rigorous complex-plane frame, mathematical solution or physical field. Their proximity to a Saturn-like form does not assert that SCM describes Saturn or that SPPT or SCM has been experimentally validated.

Files: illustrations/images/03-bent-filament-concept.png